Integrating, $\int\left(\frac{a}{1-ax}+\frac{1}{x}\right)dx = \frac{dy}{a+y}$

$$-\log(1-ax)+\log x=\log(a+y)+\log c$$

$$\log\left(\frac{x}{1-ax}\right) = \log\left(c(a+y)\right)$$

$$\left(\frac{x}{1-ax}\right) = c\,(a+y)$$

$x = (1-ax)(a+y)c$ which is the required solution.

Example 4.12

The marginal cost function of manufacturing x gloves is $6+10x-6x^2$. The total cost of producing a pair of gloves is ₹100. Find the total and average cost function.

Solution:

$$\text{Given } MC = 6+10x-6x^2$$

$$\text{i.e., } \frac{dc}{dx} = 6+10x-6x^2$$

$$dc = (6+10x-6x^2)dx$$

$$\int dc = \int(6+10x-6x^2)dx+k$$

$$c = 6x+10\frac{x^2}{2}-6\frac{x^3}{3}+k$$

$$c = 6x+5x^2-2x^3+k \quad (1)$$

Given $c = 100$ when $x = 2$

$$\therefore \quad (1) \Rightarrow 100 = 12+5(4)-2(8)+k$$

$$\Rightarrow k = 84$$

$$\therefore \quad (1) \Rightarrow c\,(x) = 6x+5x^2-2x^3+84$$

$$\text{Average Cost } AC = \frac{c}{x} = 6+5x-2x^2+\frac{84}{x}$$

Example 4.13

The normal lines to a given curve at each point (x, y) on the curve pass through the point (1,0). The curve passes through the point (1,2). Formulate the differential equation representing the problem and hence find the equation of the curve.

Solution:

Slope of the normal at any point $P(x,y) = -\frac{dx}{dy}$

Let Q be (1,0)

Slope of the normal PQ *is* $\frac{y_2-y_1}{x_2-x_1}$

i.e., $\frac{y-0}{x-1} = \frac{y}{x-1}$

$\therefore -\frac{dx}{dy} = \frac{y}{x-1} \Rightarrow \frac{dx}{dy} = \frac{y}{1-x}$, which is the differential equation

$$\text{i.e., } (1-x)dx = ydy$$

$$\int(1-x)dx = \int ydy+c$$

$$x-\frac{x^2}{2} = \frac{y^2}{2}+c \quad (1)$$

Since it passes through (1,2)

$$1-\frac{1}{2} = \frac{4}{2}+c$$

$$c = \frac{1}{2}-2 = -\frac{3}{2}$$

Put $c = \frac{-3}{2}$ in (1)

$$x-\frac{x^2}{2} = \frac{y^2}{2}-\frac{3}{2}$$

$$2x-x^2 = y^2-3$$

$$\Rightarrow y^2 = 2x-x^2+3,$$

which is the equation of the curve

Example 4.14

The sum of ₹2,000 is compounded continuously, the nominal rate of interest being 5% per annum. In how many years will the amount be double the original principal? $(\log_e 2 = 0.6931)$

Solution:

Let P be the principal at time 't'

$$\frac{dP}{dt} = \frac{5}{100}P = 0.05P$$

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